

Do System Prompts Leave Behavioral Fingerprints?

A Large-Scale Empirical Study of Clone Detection via Output Similarity

Linghan Chen^{1*} Yudong Gao^{2*} Jiyao Wang¹ Kaiyan Ji¹ Honglong Chen^{3†}

¹University of Adelaide ²Hong Kong University of Science and Technology

³China University of Petroleum

linghanchen2004@gmail.com

Abstract

System prompts can be extracted from commercial LLMs with over 80% success and redeployed at zero cost, yet a prompt owner has no way to verify whether a suspected deployment is a clone. We propose Black-Box Behavioral Fingerprinting (BBF): the prompt owner registers a behavioral signature from model outputs and later tests whether a suspect deployment matches that signature more closely than an unrelated baseline. BBF requires only black-box API access. Through a large-scale study (4 model families, 8 benchmarks, 288,000 responses), we find that prompt choice explains 24.4% of output variance and same-model detection reaches AUC 0.876. Cross-model performance is bounded by detector identity, with off-diagonal AUC ranging from 0.845 (Claude as detector) down to 0.665 (Qwen) and overall mean 0.725. BBF resists non-adaptive prompt paraphrasing (AUC ≥ 0.889) and is robust to imperfect extraction, but a single-sentence formal-tone prefix can collapse detection on short structured outputs (MNL 0.978 $\rightarrow$ 0.547), isolating style-invariant detection as the key open problem. Diagnostic Query Optimization, a zero-cost query selection rule, adds +0.120 to cross-model AUC.

1 Introduction

LLM application platforms such as OpenAI’s GPT Store, Coze, and Dify have given rise to a new class of intellectual property: the system prompt. A carefully crafted prompt can transform a general-purpose model into a specialized product by encoding domain expertise, reasoning strategies, and output conventions, and LLM-based agents extend this further by encoding tool-use workflows, multi-step reasoning chains, and safety guardrails (Wang et al., 2024a). Our study covers the single-turn case directly; the multi-turn agent setting is a natural

extension we discuss in Limitations. Such prompt-based applications are now widely deployed commercially, and the prompt is often a key differentiator between competing products. However, unlike model weights, which require significant compute to replicate, a system prompt can be extracted verbatim through adversarial queries, with success rates exceeding 80% on commercial systems (Hui et al., 2024; Sha and Zhang, 2024). Once extracted, a prompt can be redeployed on any model, on any platform, at zero cost. Yao et al. (2025) showed that this entire pipeline, from extraction to cloning to deployment, can be fully automated against real-world prompt services.

Existing defenses focus on *preventing* extraction (Schulhoff et al., 2023; Chen et al., 2025; Das and Amini, 2025), yet the extraction success rates above show that prevention alone is not yet reliable. A single successful extraction also permanently compromises the prompt, since the extracted text can be copied and redistributed without limit. We therefore argue for a complementary approach: *detecting* whether a suspected deployment is running a stolen prompt. Just as copyright enforcement provides recourse even when access control fails, post-extraction detection offers recourse even after the prompt has already been stolen.

Whether such detection is feasible hinges on a prior question: does a system prompt leave a usable trace in model outputs at all? We find that it does, and strongly: prompt choice explains 24.4% of output-embedding variance across eight benchmarks ($\eta^2 = 0.244$, a large effect (Cohen, 1988); Section 5), and clone outputs sit closer to the original than unrelated-prompt outputs do in 118 of 128 experimental scenarios (Figure 1). This imprint is what makes post-extraction detection feasible in principle; the open questions concern its *robustness*. Does it survive *cross-model transfer* to a different LLM with different tokenization and decoding? Does it survive an adversary who *paraphrases* the

*Equal contribution.

†Corresponding author.

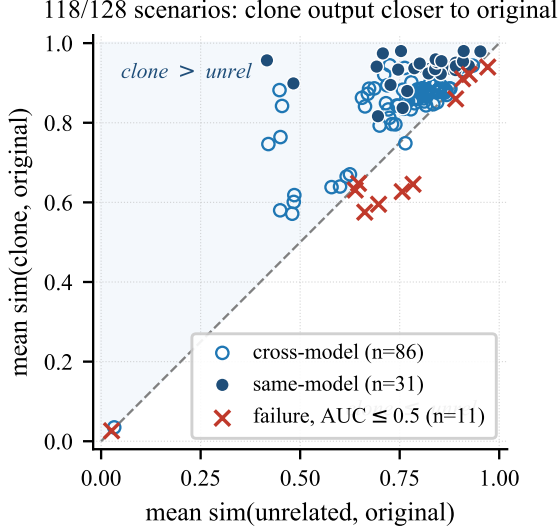


Figure 1: Each point is one of 128 scenarios (8 benchmarks $\times$ 4 registration $\times$ 4 detection models). Red $\times$ marks failures ($\text{AUC} \leq 0.5$).

stolen prompt? Which *domains and model pairs* are boundary conditions, and how many *diagnostic queries* does a defender need?

We propose Black-Box Behavioral Fingerprinting (BBF), inspired by content fingerprinting in multimedia: the prompt owner registers a behavioral signature from outputs of the original deployment, and later tests whether a suspect produces outputs closer to that signature than to an unrelated baseline. BBF needs only black-box API access.

Through a large-scale empirical study (128 scenarios, 4 model families, 8 benchmarks, 288,000 responses), we make three contributions. First, we map when output-similarity detection succeeds and when it fails across the full $4 \times 4 \times 8$ space: same-model $\text{AUC} = 0.876$, cross-model AUC ranges 0.845 (Claude as detector) to 0.665 (Qwen as detector), and 11 of 128 scenarios fail along three identifiable patterns (short outputs, data artifacts, model-pair style mismatch; Section 6). Second, we propose Diagnostic Query Optimization (DQO), a prompt-free and transferable query selection rule with mean held-out gains of +0.120 cross-model and +0.045 same-model (Section 3.4). Third, across Sections 7–8, we show that BBF survives heavy prompt paraphrasing ($\text{AUC} \geq 0.889$), characterize one adaptive attack that does break it on short-output tasks, and establish that 25 diagnostic queries already suffice for reliable detection.

BBF is a *verification* tool, not a discovery tool: it confirms or refutes suspicion about a specific

deployment and is intended as a complement to existing IP enforcement, not a surveillance method.

2 Related Work

System prompt extraction. System prompts can be extracted through adversarial queries (Schulhoff et al., 2023; Hui et al., 2024; Sha and Zhang, 2024), indirect prompt injection (Greshake et al., 2023; Toyer et al., 2024), and automated attacks against real-world platforms (Yao et al., 2025). Defenses include proxy architectures (Chen et al., 2025) and various filtering strategies surveyed by Das and Amini (2025). All of these focus on *preventing* extraction, whereas our work targets *post-extraction detection* as a complementary defense layer.

Watermarking and model fingerprinting.

LLM watermarking (Kirchenbauer et al., 2023; Christ et al., 2024) embeds a signal into model outputs during generation, requiring modification of the decoding process and model provider cooperation. Model fingerprinting (Xu et al., 2024; Zeng et al., 2024; Sun et al., 2025; Li et al., 2023a) identifies which *model* produced an output, typically requiring weight access or fine-tuning; prompts have also been used as units of personalization in adjacent privacy contexts such as federated unlearning (Wu et al., 2025). Our work differs along two axes: we detect whether a specific *system prompt* has been cloned (not which model is used), and we require only black-box API access with no weights, no generation modification, and no platform cooperation. Direct numerical comparison against these methods is not meaningful given the different access assumptions; the closest output-only baseline is lexical similarity (TF-IDF / Jaccard), which we benchmark in Section 7.2.

Sentence embeddings. BBF builds on sentence-level representations (Reimers and Gurevych, 2019; Li et al., 2023b; Wang et al., 2024b) as benchmarked by MTEB (Muennighoff et al., 2023). These embeddings serve as a measurement tool; our contribution is establishing when the resulting similarity signal suffices for clone detection. A natural alternative would be to ask a strong LLM to judge whether two output sets share a prompt (LLM-as-judge); we did not adopt this baseline because each detection call would invoke a frontier model on the full diagnostic set (cost scales with $K \cdot N$), the judgments are themselves prompt-

conditioned and non-deterministic, and embedding similarity gives a closed-form, reproducible decision rule. A direct comparison against an LLM-as-judge baseline is a useful next experiment.

3 Method

3.1 Problem Formulation

Let $\mathcal{P}$ be a protected system prompt and $\mathcal{M}$ an LLM. For query q , the model produces a stochastic output $o \sim \mathcal{M}(q; \mathcal{P})$. A sentence encoder $\phi : \mathcal{O} \rightarrow \mathbb{R}^d$ maps outputs into a normalized embedding space. The prompt $\mathcal{P}$ thus induces a query-conditional distribution $p_{\mathcal{P}}(\cdot | q)$ over $\mathbb{R}^d$. An adversary extracts $\mathcal{P}$, creates a clone $\mathcal{P}'$, and deploys it on a potentially different model $\mathcal{M}'$. The clone detection problem is: given only black-box API access to both deployments, determine whether the behavioral distribution $p_{\mathcal{P}'}(\cdot | q)$ is closer to $p_{\mathcal{P}}(\cdot | q)$ than to that of an unrelated prompt $\mathcal{P}_u$.

Two prompts are *functionally equivalent* if they elicit comparable task behavior; the clones we evaluate are semantic paraphrases of their originals that we manually verified to preserve task instructions on spot-checked benchmarks. BBF operationalizes detection of this equivalence as a centroid-closeness test in embedding space, comparing the suspect’s centroid against the original and an unrelated same-domain baseline.

3.2 Threat Model

The detection setting involves two parties with asymmetric capabilities. The *defender* (prompt owner) can query the suspected clone’s public API, send ≥ 25 diagnostic queries, and compute a baseline using one unrelated prompt from the same domain. The defender does not need to know which model the suspect uses, does not need model weights or logits, and does not need cooperation from the suspect’s platform. One assumption is implicit: the defender must know the protected prompt’s broad domain in order to construct a same-domain unrelated baseline. This is mild in practice (a prompt owner who suspects cloning typically knows what their own prompt does), but it is a real prior, and misspecifying the domain weakens the baseline. Section 8.3 reports a leave-one-domain-out cross-validation in which the baseline is constructed from a different benchmark, yielding mean AUC 0.793 versus 0.876 in-domain, which quantifies the cost of imperfect domain knowledge. The *adversary* may deploy the stolen prompt on any

model family, may paraphrase the prompt to obscure its origin (evaluated in Section 8), and may use prompts obtained through imperfect extraction rather than verbatim copies (evaluated in Section 8.3). This asymmetry, where the defender has minimal access while the adversary has maximal freedom, makes the detection problem challenging and motivates a method that relies on behavioral signals rather than direct comparison of the prompt text.

3.3 BBF: Black-Box Behavioral Fingerprinting

BBF is intentionally simple: it casts clone detection as a two-sample comparison in embedding space, using only off-the-shelf sentence embeddings and cosine similarity. This design choice isolates the core empirical question (do behavioral fingerprints exist and transfer?) from confounds that a more complex method would introduce. BBF operationalizes detection through two phases.

Registration. For diagnostic queries $\mathcal{Q} = \{q_1, \dots, q_K\}$, the defender draws N i.i.d. output samples from the original deployment and estimates the behavioral centroid μ_k for each query:

$$\hat{\mu}_k = \frac{1}{N} \sum_{i=1}^N \phi(\mathcal{M}(q_k; \mathcal{P})_i). \quad (1)$$

The fingerprint $\mathcal{F} = \{\hat{\mu}_1, \dots, \hat{\mu}_K\}$ is a set of centroid estimates, one per query, that characterize $p_{\mathcal{P}}(\cdot | q_k)$. Averaging over N samples reduces the variance introduced by stochastic decoding (temperature > 0). Section 7.1 shows that $K = 25$ queries with $N = 3$ samples per query are sufficient for our experiments.

Detection. Given a suspected deployment using prompt $\mathcal{P}^*$ on model $\mathcal{M}'$, the defender estimates suspect centroids $\hat{\mu}_k^*$ in the same manner and computes the query-level similarity:

$$s_k^{\text{clone}} = \cos(\hat{\mu}_k, \hat{\mu}_k^*). \quad (2)$$

Raw similarity is confounded by query content: for any factual question, outputs from different prompts share substantial semantic overlap simply because they answer the same question. To control for this, the defender also collects outputs under an unrelated prompt $\mathcal{P}_u$ from the same domain and computes $s_k^{\text{unrel}} = \cos(\hat{\mu}_k, \hat{\mu}_k^{\text{unrel}})$. The detection statistic is the *controlled gap*:

$$\delta_k = s_k^{\text{clone}} - s_k^{\text{unrel}}. \quad (3)$$

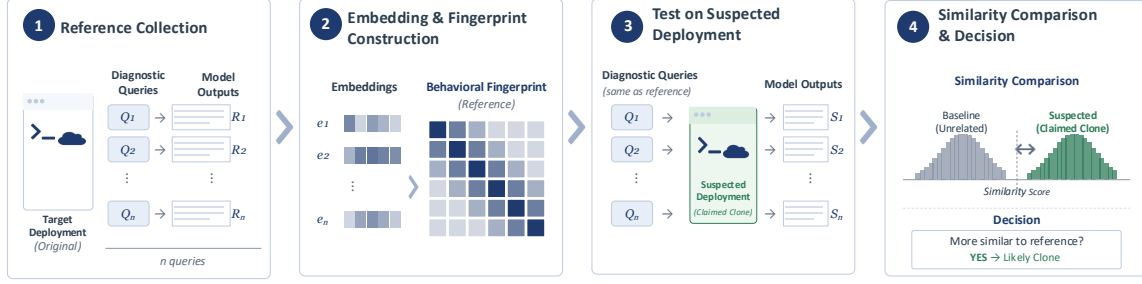


Figure 2: BBF at a glance. **(1)** The defender sends n diagnostic queries to the original deployment and collects responses. **(2)** Responses are embedded and aggregated into a behavioral fingerprint that characterizes the original prompt’s output distribution. **(3)** The same queries are sent to the suspected deployment. **(4)** The suspect’s outputs are compared in embedding space to both the reference fingerprint and an unrelated-prompt baseline; greater similarity to the reference indicates cloning. Formal definitions in Section 3.

Under the null hypothesis H_0 (the suspect uses a prompt unrelated to $\mathcal{P}$), the expected gap $\mathbb{E}[\delta_k] \approx 0$. Under the alternative H_1 (the suspect uses a clone of $\mathcal{P}$), $\mathbb{E}[\delta_k] > 0$ because functional equivalence implies $s_k^{\text{clone}} > s_k^{\text{unrel}}$.

Aggregation. Individual queries provide noisy evidence. For evaluation, we report AUC over all K queries, i.e., $\Pr(\delta_k^{\text{clone}} > \delta_j^{\text{unrel}})$ for randomly drawn k, j . For deployment, the decision rule aggregates via a *majority vote*: send T queries, compute $\hat{r} = \frac{1}{T} \sum_{k=1}^T \mathbf{1}[\delta_k > 0]$, and declare cloning if $\hat{r} > 0.5$. The proposition below shows that this aggregation yields exponentially increasing confidence as the query budget grows.

Proposition 1 (Detection guarantee) *Let $p = \Pr(\delta_k > 0 \mid H_1) > 1/2$ be the per-query detection probability under the cloning hypothesis, and assume queries are independent. The majority vote detector over T queries achieves scenario-level detection probability*

$$P_{\text{detect}}(T, p) \geq 1 - \exp(-2T(p - \frac{1}{2})^2). \quad (4)$$

A short Hoeffding-style proof is in Appendix A.2. For $p = 0.786$ (the empirical query-level recall) and $T = 25$, this bound gives $P_{\text{detect}} \geq 1 - e^{-4.08} > 0.983$. The exact binomial calculation yields $P_{\text{detect}} > 0.999$, and this matches the scenario-level detection probability we observe empirically in non-failure regimes (Section 8.3), so the bound, while derived under independence, is not noticeably tight in practice for our query budget. The independence assumption is an approximation: queries from the same benchmark exhibit some positive correlation, which would make a tighter bound depend on the correlation

structure. Nevertheless, exponential convergence in T keeps detection reliable under moderate correlation given a sufficient query budget. The bound is informative when p stays well above 0.5; in failure-mode scenarios (Section 6.4) where p approaches 0.5, reaching equivalent detection confidence calls for a query budget considerably larger than the $T = 25$ default used in our main experiments.

3.4 Diagnostic Query Optimization

Not all queries carry equal diagnostic value. For a query q_k with a canonical answer, $p_{\mathcal{P}}(\cdot \mid q_k) \approx p_{\mathcal{P}_u}(\cdot \mid q_k)$, so $\delta_k \approx 0$ regardless of whether cloning has occurred. Including such low-information queries in $\mathcal{Q}$ dilutes detection power. Diagnostic Query Optimization (DQO) addresses this by selecting queries that maximize *prompt sensitivity* prior to any interaction with the suspected clone, with empirical validation of the selection rule presented in Section 7.1 below.

Define the prompt sensitivity of query q_k as the divergence between the original and unrelated behavioral distributions. Since we do not have access to these distributions directly, DQO uses a tractable proxy: the baseline similarity $s_k^{\text{unrel}} = \cos(\hat{\mu}_k, \hat{\mu}_k^{\text{unrel}})$. Low s_k^{unrel} indicates that the original and unrelated prompts produce highly divergent outputs for q_k (high prompt sensitivity); high s_k^{unrel} indicates convergent outputs where prompt variation has little effect. The optimized diagnostic set is:

$$\mathcal{Q}^* = \arg \min_{\mathcal{S} \subset \mathcal{Q}_{\text{cand}}, |\mathcal{S}|=T} \sum_{q_k \in \mathcal{S}} s_k^{\text{unrel}}, \quad (5)$$

DQO thus reduces to picking, from the candidate pool, the T queries with the smallest baseline similarity to the unrelated prompt. Algorithmically,

DQO is a one-shot ranking heuristic rather than a learned model; its contribution is operational (it is *prompt-free* and *transferable*, with the same ranked set reusable across suspected deployments) rather than methodological.

The following proposition provides a sufficient condition under which DQO is optimal.

Proposition 2 (DQO optimality) *If the expected gap $\mathbb{E}[\delta_k \mid H_1]$ is a monotonically decreasing function of baseline similarity s_k^{unrel} , then the DQO selection rule (Eq. 5) maximizes the expected total detection signal $\sum_{q_k \in Q^*} \mathbb{E}[\delta_k \mid H_1]$ over all subsets of size T .*

A short greedy-argument proof is in Appendix A.2. The monotonicity assumption is empirically supported: across 32,000 query-scenario pairs, the Pearson correlation between s_k^{unrel} and δ_k is $r = -0.516$ ($p \approx 0$; Section 7.1). This scoring function is also *prompt-free* (it depends only on the defender’s own prompts, requiring no access to the suspected clone) and *transferable* (the same ranked query set can be reused across different suspected deployments). Validation on held-out data confirms its effectiveness: the selection rule learned on 125 MMLU training queries lifts AUC from 0.878 to 0.980 on 125 held-out queries (Section 7.1); Algorithm 1 in Appendix A gives the full BBF procedure.

4 Experimental Setup

Our evaluation spans four model families and eight benchmarks across 288,000 generated responses.

Models. The four model families are Claude (claude-haiku-4-5, Anthropic), GPT (gpt-4o-mini, OpenAI), DeepSeek (DeepSeek-V3, via Silicon-Flow), and Qwen (qwen-plus, Alibaba). These span four widely-deployed commercial APIs from distinct providers, including both closed-source and open-weight model families.

Benchmarks. We evaluate on 8 benchmarks spanning 7 domains (Table 1). Diagnostic queries are drawn directly from standard NLP datasets without adversarial crafting or domain-specific engineering. For each benchmark, we design three system prompts: an *original* (the protected prompt), a *clone* (semantic paraphrase preserving functionality), and an *unrelated* (same broad domain, different task). With 4 models forming a 4×4 cross-model matrix, this yields 16 scenarios per benchmark and 128 total scenarios.

Domain	Benchmark	Queries
General QA	MMLU (57 subjects)	250
	TriviaQA	250
Classification	SST-2 (sentiment)	250
NLI	MNLI (inference)	250
Medical	MedQA	250
Legal	CUAD (contracts)	250
Math	GSM8K	250
Summarization	CNN/DailyMail	250
Total		2,000

Table 1: Benchmarks used for evaluation. Each benchmark uses 250 queries sampled from standard NLP datasets.

Sampling and evaluation. We use $N = 3$ samples per query, temperature 0.7, and max 400 output tokens. Each experiment generates $250 \times 3 \times 3 = 2,250$ model responses (original, clone, and unrelated prompts $\times$ 250 queries $\times$ 3 samples), totaling 288,000 responses across all 128 scenarios. For each query we compare clone similarity to the unrelated baseline, and report AUC over all queries.

Embedding model. All embeddings use all-MiniLM-L6-v2 (Reimers and Gurevych, 2019) (384 dimensions, 22M parameters). Ablation across three architectures (MiniLM-L6, BGE-large, GTE-large) confirms a mean AUC difference below 0.02, as detailed in Section 7.2.

Unrelated prompt design. For each benchmark, the unrelated prompt shares the same broad domain as the original (e.g., both are medical) but specifies a different task (e.g., diagnosis vs. drug interaction). This same-domain design is more conservative than a cross-domain baseline, setting a higher bar for detection by controlling for domain-level similarity.

5 The Behavioral Imprint of System Prompts

Section 1 previewed our central observation: system prompt choice has a large, measurable effect on model outputs, and this is what licenses output similarity as a detection signal. We quantify that effect via a one-way ANOVA in which prompt identity (original / clone / unrelated) is the factor and output-embedding similarity is the dependent variable, computing η^2 within each benchmark to avoid cross-benchmark variance inflation (Figure 3).

The mean $\eta^2 = 0.244$ constitutes a large effect by Cohen’s standard (Cohen, 1988), whose thresholds for η^2 are 0.01 (small), 0.06 (medium),

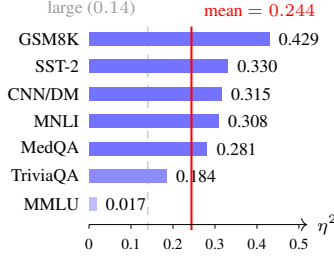


Figure 3: Within-benchmark η^2 effect sizes (CUAD excluded; see Section 6.4). Dashed line: Cohen’s large-effect threshold ($\eta^2 > 0.14$).

and 0.14 (large); 0.244 is roughly $1.7\times$ the large-effect threshold and means that prompt choice explains on average 24.4% of within-benchmark output variance. The effect varies substantially across domains: GSM8K shows the strongest prompt influence ($\eta^2 = 0.429$), consistent with the diversity of step-by-step reasoning styles that different prompts elicit, while MMLU shows a small effect ($\eta^2 = 0.017$), reflecting its many canonically-answered questions where any prompt produces similar responses. CUAD is excluded here due to a HuggingFace dataset issue (Section 6.4) that introduced a prompt-query domain mismatch; AUC-based analyses elsewhere in the paper are unaffected by this artifact.

This establishes prompt as a load-bearing factor in output variance, a necessary but not sufficient condition for detection; whether the signal actually separates clone from unrelated outputs is the empirical question Section 6 addresses next.

6 Main Results

Having established that system prompts leave measurable behavioral fingerprints, we now evaluate whether this signal is sufficient for practical clone detection. We report results across all 128 scenarios, organized by same-model detection, cross-model transfer, and failure modes.

6.1 Overview

Table 2 summarizes all 128 scenarios: overall mean AUC 0.763 and same-model 0.876. Cross-model performance is governed by which model serves as detector: off-diagonal column means span 0.845 (Claude) to 0.665 (Qwen), with overall cross-model mean 0.725 (Section 6.3 and Table 3). The 11 failures ($\text{AUC} \leq 0.50$) concentrate in CUAD (5), SST-2 (3), and CNN/DM (3); excluding CUAD’s data-loading issue (Section 6.4), the method-attributable

Benchmark	n	Mean	Min	Max	Fails
MMLU	16	0.720	0.500	0.918	0
TriviaQA	16	0.796	0.676	0.944	0
SST-2	16	0.775	0.239	1.000	3
MNLI	16	0.823	0.589	0.983	0
MedQA	16	0.798	0.529	0.999	0
CUAD [†]	16	0.651	0.235	0.987	5
GSM8K	16	0.764	0.544	0.977	0
CNN/DM	16	0.773	0.449	1.000	3
All	128	0.763	0.235	1.000	11
Excl. CUAD	112	0.779	0.239	1.000	6

Table 2: Per-benchmark AUC summary. n = scenarios; “Fails” = $\text{AUC} \leq 0.50$. [†]CUAD results are affected by a dataset loading issue (see Section 6.4); the “Excl. CUAD” row provides method-only failure rates (6/112 = 5.4%).

Reg.	Detection Model				Mean
	Claude	GPT	DS	Qwen	
Claude	.967	.693	.688	.619	.742
GPT	.830	.875	.719	.736	.790
DeepSeek	.863	.782	.867	.640	.788
Qwen	.843	.649	.636	.796	.731
Col. mean	.876	.750	.728	.698	.763

Table 3: Mean AUC across 8 benchmarks for each registration $\times$ detection model pair. Bold diagonal = same-model. Column means reveal that *detection* model identity matters most ($R^2 = 0.161$), exceeding benchmark ($R^2 = 0.096$) and registration model ($R^2 = 0.050$).

failure rate is 6 of 112 (5.4%).

6.2 Same-Model Detection

When the fingerprint is registered and the suspect runs on the same model family, BBF operates in its most common detection setting. Same-model detection is reliable, with an overall mean AUC of 0.876 across 32 scenarios (per-model means: Claude 0.967, GPT 0.875, DeepSeek 0.867, Qwen 0.796). Claude stands out for its consistency (6 of 8 benchmarks exceed 0.94). Only one same-model scenario fails (CUAD on GPT→GPT, 0.448), discussed in Section 6.4. The full 4×8 benchmark-by-model breakdown is in Appendix B.1 (Table 6).

6.3 Cross-Model Detection

In practice, an adversary deploys the stolen prompt on a different model. Restricted to the 12 off-diagonal cross-model pairs in Table 3, mean AUC ranges from 0.845 with Claude as detector down to 0.665 with Qwen. Detector-model identity accounts for $R^2 = 0.161$ of AUC variance across the full matrix, dominating the $R^2 = 0.050$ con-

tributed by registration choice.

Transfer works because prompt-specific patterns (reasoning steps, caveats, structure) are partially model-agnostic and captured by sentence embeddings. Gaps are sometimes *larger* cross-model than same-model (e.g., Claude→GPT +0.101 vs. Claude→Claude +0.093 on MMLU), because unrelated prompts diverge more across families.

6.4 Failure Modes

Of 128 scenarios, 11 produce $\text{AUC} \leq 0.50$. The failures cluster into three distinct patterns:

Short-output directional failures (SST-2, 3 failures). Claude-registered fingerprints fail to transfer outward ($C \rightarrow G = 0.239$, $C \rightarrow D = 0.264$), and Qwen→GPT fails (0.495). Sentiment outputs are extremely short (often <50 tokens), and Claude produces idiosyncratic concise formats. The failure is directional: GPT and DeepSeek fingerprints transfer effectively. Output above 50 tokens is therefore a necessary precondition for reliable transfer across model families.

Data artifact failures (CUAD, 5 failures). CUAD contributes 5 failures including the sole same-model failure ($GG = 0.448$). These are *data artifacts*, not method failures: the HuggingFace cuad loader silently fell back to a SQuAD-formatted split when the expected contract-clause configuration was unavailable, so our contract-analysis prompts were issued against reading-comprehension passages. The mismatch suppresses the prompt-conditional signal because outputs collapse to passage-grounded extraction regardless of system prompt. The remaining seven benchmarks load from standard, stable splits (MMLU, TriviaQA, SST-2, MNLI, MedQA, GSM8K, CNN/DM) and exhibit no analogous prompt-query domain mismatch in the queries we sampled. Excluding CUAD, the method-attributable failure rate drops to 6 of 112 scenarios, or 5.4%.

Cross-model style mismatch (CNN/DM, 3 failures). DeepSeek→Qwen (0.500), Qwen→GPT (0.449), and Qwen→DeepSeek (0.474) reflect a systematic style mismatch between these pairs (also seen in MMLU $D \rightarrow Q = 0.500$). Both models transfer well to Claude ($D \rightarrow C$, $Q \rightarrow C > 0.85$), confirming the failure is pair-specific.

Bench.	Orig.	L1	L2	L3
MMLU	.862	.966	.922	.889
SST-2	.996	.984	1.00	.969
MNLI	.978	.997	.984	.991
MedQA	.933	.983	.958	.956
GSM8K	.991	.997	.998	1.00
CNN/DM	1.00	1.00	1.00	1.00

Table 4: Paraphrase attack (Claude→Claude): even under L3, all $\text{AUC} \geq 0.889$. The companion table on LLM-simulated extraction quality is in Appendix B.3 (Table 8).

7 Ablation Studies

Section 7.1 ablates the query selection rule and the query budget K , and Section 7.2 compares the embedding model against alternative encoders and against lexical similarity baselines.

7.1 Query Discriminability and DQO Validation

Some queries yield non-positive gaps because their content is canonical enough that any prompt produces similar output (Pearson $r = -0.516$ between s_k^{unrel} and δ_k , $p \approx 0$, $n = 32,000$). As Figure 4a shows, low-baseline queries yield the largest gaps (mean +0.199 for $s^{\text{unrel}} < 0.7$ vs. +0.011 for $s^{\text{unrel}} > 0.85$), motivating DQO (Section 3.4). DQO improves AUC in all 9 validated settings (Figure 4c), with larger gains in cross-model scenarios. A query budget of $K = 25$ already suffices for reliable detection (Figure 4b).

7.2 Embedding Model and Baseline Comparison

Detection is consistent across embedding backends: mean AUC difference is below 0.02 across MiniLM-L6, BGE-large, and GTE-large. Lexical baselines (TF-IDF, bigram Jaccard) are competitive on same-model pairs but degrade on cross-model pairs (e.g., MNLI Claude→GPT: BBF 0.893 vs. TF-IDF 0.772), confirming BBF’s advantage where surface similarity breaks down. Full per-benchmark numbers are in Appendix B.2 (Table 7).

8 Robustness

Section 8.1 evaluates paraphrase attacks; Section 8.2 evaluates adaptive prefix injection; Section 8.3 covers practical deployment conditions.

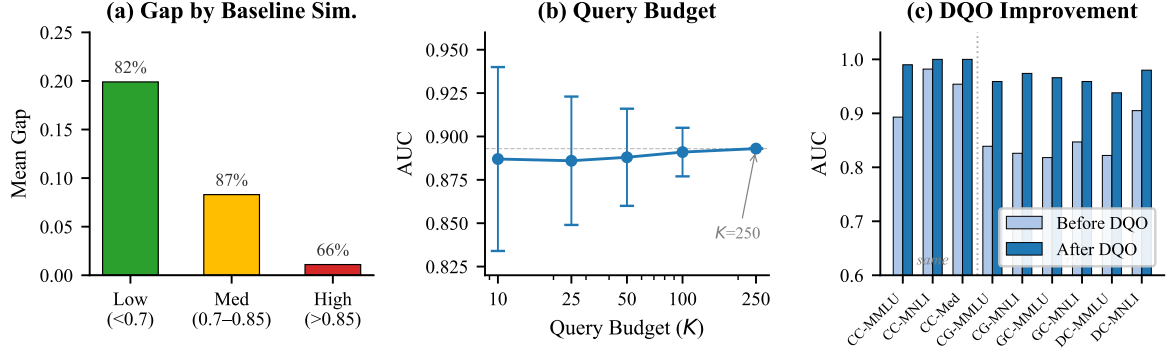


Figure 4: Query selection analysis. (a) Detection gap by baseline-similarity bin; percentages mark the proportion of positive gaps. (b) Query budget ablation on MMLU Claude→Claude. (c) DQO before/after across 9 settings.

Prefix Strategy	MMLU	MNLI
None (baseline)	0.855	0.978
Chain-of-thought	0.792	0.958
Formal academic tone	0.863	0.547
Random padding	0.838	0.771

Table 5: Adaptive prefix injection (Claude→Claude, 50 queries). Formal tone evades MNLI detection (0.978 → 0.547) but barely shifts MMLU, which holds at 0.863.

8.1 Non-Adaptive Paraphrase Attack

GPT-4o-mini generates rewrites at three intensity levels: L1 (synonym substitution), L2 (structural reorganization), and L3 (complete functional rewrite), evaluated on Claude→Claude across 6 benchmarks (50 queries each). Even under L3, all AUC values remain ≥ 0.889 (Table 4). L1 paraphrase sometimes raises AUC above the original clone: MMLU rises from 0.862 to 0.966, suggesting that light rewrites can smooth stylistic noise.

8.2 Adaptive Attack: Prefix Injection

Paraphrasing the prompt fails to evade BBF (§8.1), because the embedding signature tracks behavior rather than wording. An adversary who instead targets the *output style*, by prepending a short instruction that overrides the prompt’s stylistic surface, poses a stronger threat. We evaluate four such prefix-injection strategies on MMLU and MNLI (Claude→Claude, 50 queries).

Table 5 reveals an asymmetry: Non-adaptive paraphrasing does not evade BBF, but targeted style-shifting can degrade detection on short, structured outputs (MNLI: 0.978 → 0.547 under formal tone) while leaving content-rich outputs unaffected (MMLU: 0.863). Defending against style-shifting

on short outputs remains open.

8.3 Deployment Conditions

We evaluate three practical deployment factors. **Imperfect extraction:** On MMLU (Claude→Claude), author-constructed clones at all five fidelity levels achieve $\text{AUC} \geq 0.879$, and LLM-simulated clones match or exceed the handcrafted baseline ($\text{AUC } 0.886 \rightarrow 0.982$; Appendix Table 8). The direction of this result is counter-intuitive: *noisier* extracted clones are at least as detectable as carefully handcrafted paraphrases. Our hypothesis is that extracted text preserves the original prompt’s functional framing and surface phrasing even when imperfect, whereas a handcrafted paraphrase, written by an independent author with their own style, injects orthogonal stylistic variance that the embedding picks up as noise. The practical implication is favorable: the realistic threat model, in which an adversary deploys text obtained from automated extraction (Hui et al., 2024; Yao et al., 2025) rather than rewriting the prompt from scratch, is precisely the regime where BBF is strongest. **Cross-domain generalization:** LOOCV across the 8 NLP benchmarks yields mean $\text{AUC} = 0.793$, with CUAD remaining the lowest at 0.651; a supplementary check on four enterprise domains (legal, financial, clinical, software architecture) confirms positive signal (mean gap +0.194, 15/16 significant; Appendix C). **Detection thresholds:** query-level recall at gap > 0 is 78.6% across 32,000 pairs; majority vote over $T = 25$ then lifts the scenario-level detection probability above 0.999.

9 Conclusion

BBF detects cloned system prompts from black-box outputs at AUC 0.876 same-model and 0.725 mean cross-model, with detector identity (0.845 Claude to 0.665 Qwen) as the dominant transfer factor. The signal is reliable for content-rich outputs and Claude or DeepSeek detectors, but collapses under output-style adaptation on short structured tasks (Section 8.2); closing that style-invariance gap is the key direction this work leaves open. Within these limits, BBF gives a prompt owner a defensible verification protocol that requires only black-box API access, no platform cooperation, and a fixed budget of 25 diagnostic queries.

Limitations

The strongest open issue is robustness to *adaptive* attacks. Section 8.2 shows that a single-sentence formal-tone prefix collapses MNLi detection from 0.978 to 0.547, and our evaluation covers only four prefix-injection strategies on two benchmarks; sandwich constructions, suffix steering, output-format steering, and multi-turn dilution remain unexplored. Style-invariant detection on short, structured outputs is therefore the natural next direction, and BBF should not be deployed against adversaries assumed to be aware of the detector. A second limitation is scope: BBF is evaluated on 8 text-based NLP benchmarks and four mid-tier commercial APIs, so generalization to multimodal tasks, real-time agents, open-source or frontier-scale models, and sandwich/RAG deployments is not established. Our evaluation is also single-turn and does not cover the agent / tool-use settings motivated in Section 1. Finally, LOOCV gives feasible but imperfect cross-domain transfer (mean AUC = 0.793), so domain-specific calibration may be needed.

Ethics Statement

BBF provides statistical evidence of prompt cloning, not legal conclusions; the legal status of system prompt cloning varies by jurisdiction. We recommend it be used only with reasonable prior suspicion of cloning, not for speculative surveillance. Diagnostic queries are drawn from established public NLP benchmarks; code and data will be released with the camera-ready version.

References

- Zhifeng Chen and 1 others. 2025. ProxyPrompt: Securing system prompts against prompt extraction attacks. *arXiv preprint arXiv:2505.11459*.
- Miranda Christ, Sam Gunn, and Or Zamir. 2024. Undetectable watermarks for language models. In *Conference on Learning Theory (COLT)*.
- Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences*, 2nd edition. Lawrence Erlbaum Associates.
- Gourab Das and Sonia Amini. 2025. System prompt extraction attacks and defenses in large language models. *arXiv preprint arXiv:2505.23817*.
- Yudong Gao, Linghan Chen, Wenhan Wu, Mia Zhou, Jiyao Wang, Kaiyan Ji, Mingyu Guo, and Honglong Chen. 2026a. Bit-flip attacks on vision-language-action models: Action-decoding architecture shapes the vulnerability. *arXiv preprint arXiv:2608.15475*.
- Yudong Gao, Qingyue Wang, Yuanyuan Yuan, Ruixuan Huang, Linghan Chen, Zimo Ji, and Shuai Wang. 2026b. PathMark: Protecting intellectual property of mixture-of-expert LLMs via path watermarks. *arXiv preprint arXiv:2607.03688*.
- Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. 2023. Not what you’ve signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*.
- Qisong He, Xinmiao Huang, Jinwei Hu, Zhuoyun Li, Yi Dong, Changshun Wu, and Xiaowei Huang. 2026. Safety-constrained reinforcement learning with post-training reachability verification for robot navigation. *arXiv preprint arXiv:2605.14174*.
- Bo Hui, Haolin Yang, Neil Gong, and Yinzhi Wang. 2024. PLeak: Prompt leaking attacks against large language model applications. In *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*.
- John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. 2023. A watermark for large language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202, pages 17061–17084. PMLR.
- Yuanchun Li, Ziqi Zhang, Bingyan Liu, Ziyue Yang, and Yunxin Liu. 2023a. ModelDiff: Testing-based DNN similarity comparison for model reuse detection. In *International Symposium on Software Testing and Analysis (ISSTA)*.
- Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. 2023b. Towards general text embeddings with multi-stage contrastive learning. *arXiv preprint arXiv:2308.03281*.

- Zhuoyun Li, Boxuan Wang, Jinwei Hu, Xiaowei Huang, and Yi Dong. 2026a. FragileFlow: Spectral control of correct-but-fragile predictions for foundation model robustness. *arXiv preprint arXiv:2605.08896*.
- Zhuoyun Li, Boxuan Wang, Changshun Wu, Xiaowei Huang, and Yi Dong. 2026b. SCOPE: Sequential conformal probing for reliable OOD rejection in LLM services. *arXiv preprint arXiv:2606.21255*.
- Xutao Mao, Xiang Zheng, and Cong Wang. 2026. Agent hacks agent: Autoresearch for production-agent red-teaming. *arXiv preprint arXiv:2607.11698*.
- Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. 2023. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Sander Schulhoff, Jeremy Pinto, Anaum Khan, Louis-François Bouchard, Chenglei Si, and 1 others. 2023. Ignore this title and HackAPrompt: Exposing systemic weaknesses of LLMs through a global scale prompt hacking competition. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Zeyang Sha and Yang Zhang. 2024. Prompt stealing attacks against large language models. *arXiv preprint arXiv:2402.12959*.
- Dominik Sun and 1 others. 2025. RoFL: Robust fingerprinting of language models. *arXiv preprint arXiv:2505.12682*.
- Sam Toyer, Olivia Watkins, Ethan Adrian Menber, Justin Svegliato, Luke Bailey, Tiffany Wang, Isaac Ong, Karim Elmaaroufi, Pieter Abbeel, Trevor Darrell, Alan Ritter, and Stuart Russell. 2024. Tensor trust: Interpretable prompt injection attacks from an online game. In *International Conference on Learning Representations (ICLR)*.
- Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, and Ji-Rong Wen. 2024a. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 18(6):186345.
- Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2024b. E5: Text embeddings by weakly-supervised contrastive pre-training. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Wenhan Wu, Huanghuang Liang, Tianyu Tu, Jiawei Jiang, Chuang Hu, and Dazhao Cheng. 2025. Mimir: Data-free federated unlearning through client-specific prompt generation for personalized models. *IEEE Transactions on Mobile Computing*.
- Jiashu Xu, Fei Wang, Mingyu Derek Ma, Pang Wei Koh, Chaowei Xiao, and Muhao Chen. 2024. Instructional fingerprinting of large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3277–3306.
- Siyang Yao and 1 others. 2025. PRSA: Prompt stealing attacks against real-world prompt services. In *USENIX Security Symposium*.
- Ziyang Zeng and 1 others. 2024. A fingerprint for large language models. *arXiv preprint arXiv:2407.01235*.
- Haobo Zhang, Xutao Mao, Guangyuan Dong, Ziwei Li, Xuanbo Su, Kaijie Chen, Jing Yang, and Zheng Lin. 2026. MemMark: State-evolution attribution watermarking for agent long-term memory systems. *arXiv preprint arXiv:2605.25002*.

A Algorithm and Proofs

A.1 Algorithm

Algorithm 1 Black-Box Behavioral Fingerprinting

Require: Original prompt $\mathcal{P}$, unrelated prompt $\mathcal{P}_u$, candidate queries $\mathcal{Q}_{\text{cand}}$, budget T , samples N , encoder ϕ

Ensure: Decision: CLONE or NOT CLONE

1: // Phase 1: Registration & Query Optimization

2: **for** $q_k \in \mathcal{Q}_{\text{cand}}$ **do**

3: $\hat{\mu}_k \leftarrow \frac{1}{N} \sum_{i=1}^N \phi(\mathcal{M}(q_k; \mathcal{P})_i)$

4: $\hat{\mu}_k^{\text{unrel}} \leftarrow \frac{1}{N} \sum_{i=1}^N \phi(\mathcal{M}(q_k; \mathcal{P}_u)_i)$

5: $s_k^{\text{unrel}} \leftarrow \cos(\hat{\mu}_k, \hat{\mu}_k^{\text{unrel}})$

6: **end for**

7: $\mathcal{Q}^* \leftarrow \text{top-}T \text{ queries with lowest } s_k^{\text{unrel}} // DQO$

8: // Phase 2: Detection

9: **for** $q_k \in \mathcal{Q}^*$ **do**

10: $\hat{\mu}_k^* \leftarrow \frac{1}{N} \sum_{i=1}^N \phi(\mathcal{M}'(q_k; \mathcal{P}^*)_i)$

11: $\delta_k \leftarrow \cos(\hat{\mu}_k, \hat{\mu}_k^*) - s_k^{\text{unrel}} // \text{from Phase 1}$

12: **end for**

13: $\hat{r} \leftarrow \frac{1}{T} \sum_{k=1}^T \mathbf{1}[\delta_k > 0]$

14: **if** $\hat{r} > 0.5$ **then**

15: **return** CLONE

16: **else**

17: **return** NOT CLONE

18: **end if**

A.2 Proofs

Proposition 1 (Detection guarantee). The number of positive-gap queries $S = \sum_{k=1}^T \mathbf{1}[\delta_k > 0]$ follows $\text{Binomial}(T, p)$ with $\mathbb{E}[S] = Tp$. A majority vote fails when $S \leq T/2$, i.e., $S - Tp \leq T(1/2 - p)$. By Hoeffding’s inequality, $\Pr(S \leq T/2) \leq \exp(-2T(p - 1/2)^2)$. $\square$

Proposition 2 (DQO optimality). Under the monotonicity assumption, ordering queries by s_k^{unrel} in ascending order is equivalent to ordering by $\mathbb{E}[\delta_k \mid H_1]$ in descending order. Selecting the top- T by descending expected gap maximizes $\sum_{q_k \in \mathcal{S}} \mathbb{E}[\delta_k \mid H_1]$ by a standard greedy argument for modular functions. $\square$

B Supplementary Tables and Figures

B.1 Same-Model Detection: Full Breakdown

Benchmark	CC	GG	DD	QQ
MMLU	0.893	0.918	0.781	0.654
TriviaQA	0.944	0.944	0.735	0.838
SST-2	0.998	0.991	1.000	0.793
MNLI	0.983	0.964	0.925	0.836
MedQA	0.953	0.959	0.999	0.671
CUAD	0.987	0.448*	0.918	0.830
GSM8K	0.977	0.976	0.793	0.755
CNN/DM	1.000	0.797	0.783	0.988
Mean	0.967	0.875	0.867	0.796

Table 6: Same-model AUC. CC = Claude→Claude, GG = GPT→GPT, DD = DeepSeek→DeepSeek, QQ = Qwen→Qwen. *CUAD GG is the only same-model failure (0.448). Overall same-model mean = 0.876.

B.2 Embedding Model and Lexical Baseline Ablation

We compare MiniLM-L6 against two larger embedding models (BGE-large, GTE-large; both 1024d) and two lexical baselines (TF-IDF, bigram Jaccard).

B.3 Extraction Quality Robustness

LLM-simulated extracted clones at four fidelity levels, on MMLU and MNLI (Claude→Claude), compared to handcrafted paraphrases.

B.4 Cross-Model AUC Heatmaps

Figure 5 visualizes the 4×4 cross-model AUC matrices for each benchmark as heatmaps, with red-bordered cells marking detection failures where AUC drops to 0.50 or below.

Bench.	Pair	Embeddings			Lexical
		MiniLM	BGE	GTE	TF-IDF
MMLU	CC	.891	.945	.918	.927
	CG	.833	—	—	.794
SST-2	CC	.997	.996	.998	—
	CG	.987	.988	.983	.930
MNLI	CC	.893	—	—	.772
	CG	.893	—	—	.772
MedQA	CC	.940	.937	.946	—
GSM8K	CC	.998	.998	.994	—
CNN/DM	CC	1.00	1.00	1.00	—

Table 7: Similarity measure comparison. Embedding models: mean AUC difference < 0.02. TF-IDF: competitive on same-model (CC) but degrades on cross-model (CG) pairs.

Bench.	Clone type	AUC	Gap	$\bar{s}_{\text{cl.}}$
MMLU	Handcrafted	.886	+0.091	.937
	Extr. (good)	.935	+0.110	.951
	Extr. (partial)	.959	+0.109	.954
	Extr. (noisy)	.982	+0.121	.963
MNLI	Handcrafted	.985	+0.092	.949
	Extr. (good)	.984	+0.102	.953
	Extr. (partial)	.980	+0.096	.959
	Extr. (noisy)	1.00	+0.119	.971

Table 8: Extraction quality (Claude→Claude). LLM-simulated extracted clones are at least as detectable as handcrafted paraphrases; noisier extractions are slightly more detectable, consistent with the hypothesis discussed in Section 8.3.

C Enterprise Skill Domain Validation

To probe generalization beyond NLP benchmarks, we evaluate BBF on four enterprise system-prompt domains: legal contract analysis, equity valuation, clinical differential diagnosis, and software architecture review. Each domain uses 5 boundary queries hand-designed to maximize prompt sensitivity, with $N = 3$ samples per query, evaluated across the same 2×2 Claude/GPT same/cross-model design (16 scenarios). Across all 16, mean controlled gap is +0.194 (range +0.034 to +0.646), with 15 of 16 significant at $p < 0.05$. Same-model mean AUC is 0.96 and cross-model is 0.92; the small per-domain query budget inflates AUC ceilings, so we report gap as the primary statistic. The single non-significant scenario is software architecture under Claude→GPT transfer ($p = 0.21$), echoing the main-result observation that programming-style outputs vary more across model families. These results indicate that BBF generalizes beyond standardized NLP benchmarks to enterprise system-prompt deployments, though sample sizes warrant a larger follow-up study.

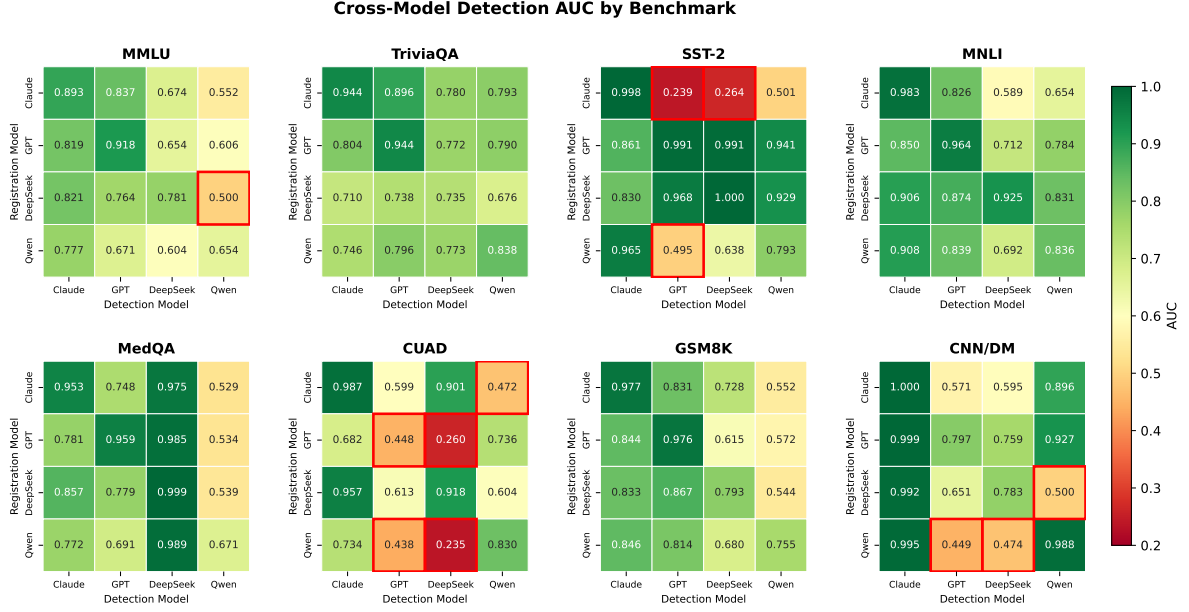


Figure 5: Cross-model detection AUC heatmaps for all 8 benchmarks. Each cell shows the AUC when registering a fingerprint on the row model and detecting on the column model. Red borders mark failures ($\text{AUC} \leq 0.50$). Failures concentrate in CUAD (5), SST-2 (3), and CNN/DM (3), primarily involving Claude-registered fingerprints detected on GPT/DeepSeek (short-output tasks) or GPT-registered fingerprints on DeepSeek (CUAD).

D System Prompts and Full Per-Scenario Results

For each benchmark, we design three prompts: an *original* (the protected system prompt), a *clone* (semantic paraphrase preserving functionality), and an *unrelated* (same broad domain but different task). Full prompt texts for all 8 benchmarks, together with per-query similarity scores and gap distributions for the 128 scenarios, are provided in the supplementary materials. Table 9 lists the 128 scenario identifiers and their AUC values.

D.1 Example Prompt Triplet (MedQA)

To make the design concrete, we reproduce the full three-prompt set used for the MedQA benchmark. The *original* and *clone* target the same skill (board-style clinical reasoning) with different surface phrasing and structure; the *unrelated* prompt stays in the medical domain but specifies a different task (curriculum design), so it controls for domain-level overlap without sharing the protected prompt’s functionality.

Original.

You are a clinical reasoning assistant for medical board-style questions. When analyzing medical questions: identify the key clinical presentation (symptoms, signs, lab values, imaging);

generate a differential diagnosis ranked by likelihood; use pathophysiology to connect findings to diagnoses; for multiple-choice, eliminate options systematically using clinical reasoning; distinguish between similar conditions (e.g., Crohn’s vs. UC, Type 1 vs. Type 2 DM); apply clinical decision rules and guidelines where applicable; note when a question tests a classic “buzzword” association. Format: (1) Key findings, (2) Differential, (3) Reasoning, (4) Answer.

Clone (semantic paraphrase).

You are a medical knowledge assistant for board exam preparation. For each clinical question: extract critical clinical data (symptoms, vitals, labs, imaging results); build a ranked differential diagnosis list; link findings to underlying pathophysiology; systematically eliminate wrong answer choices with clinical logic; differentiate look-alike conditions using distinguishing features; apply relevant clinical guidelines and scoring systems; recognize classic presentation patterns and associations. Format: (1) Clinical clues, (2) Differentials, (3) Analysis, (4) Best answer.

Unrelated (same domain, different task).

You are a medical education curriculum designer. Design case-based learning modules for medical students; create

standardized patient encounter scripts;
develop OSCE station checklists with
scoring rubrics; map learning objectives
to medical competency frameworks;
recommend simulation scenarios for
procedural skills training. Format: (1)
Learning objectives, (2) Case outline,
(3) Assessment rubric.

#	Bench.	Scenario	AUC	#	Bench.	Scenario	AUC
1	MMLU	C→C	0.893	65	MedQA	C→C	0.953
2	MMLU	C→G	0.837	66	MedQA	C→G	0.748
3	MMLU	C→D	0.674	67	MedQA	C→D	0.975
4	MMLU	C→Q	0.552	68	MedQA	C→Q	0.529
5	MMLU	G→C	0.819	69	MedQA	G→C	0.781
6	MMLU	G→G	0.918	70	MedQA	G→G	0.959
7	MMLU	G→D	0.654	71	MedQA	G→D	0.985
8	MMLU	G→Q	0.606	72	MedQA	G→Q	0.534
9	MMLU	D→C	0.821	73	MedQA	D→C	0.857
10	MMLU	D→G	0.764	74	MedQA	D→G	0.779
11	MMLU	D→D	0.781	75	MedQA	D→D	0.999
12	MMLU	D→Q	0.500	76	MedQA	D→Q	0.539
13	MMLU	Q→C	0.777	77	MedQA	Q→C	0.772
14	MMLU	Q→G	0.671	78	MedQA	Q→G	0.691
15	MMLU	Q→D	0.604	79	MedQA	Q→D	0.989
16	MMLU	Q→Q	0.654	80	MedQA	Q→Q	0.671
17	TriviaQA	C→C	0.944	81	CUAD	C→C	0.987
18	TriviaQA	C→G	0.896	82	CUAD	C→G	0.599
19	TriviaQA	C→D	0.780	83	CUAD	C→D	0.901
20	TriviaQA	C→Q	0.793	84	CUAD	C→Q	0.472*
21	TriviaQA	G→C	0.804	85	CUAD	G→C	0.682
22	TriviaQA	G→G	0.944	86	CUAD	G→G	0.448*
23	TriviaQA	G→D	0.772	87	CUAD	G→D	0.260*
24	TriviaQA	G→Q	0.790	88	CUAD	G→Q	0.736
25	TriviaQA	D→C	0.710	89	CUAD	D→C	0.957
26	TriviaQA	D→G	0.738	90	CUAD	D→G	0.613
27	TriviaQA	D→D	0.735	91	CUAD	D→D	0.918
28	TriviaQA	D→Q	0.676	92	CUAD	D→Q	0.604
29	TriviaQA	Q→C	0.746	93	CUAD	Q→C	0.734
30	TriviaQA	Q→G	0.796	94	CUAD	Q→G	0.438*
31	TriviaQA	Q→D	0.773	95	CUAD	Q→D	0.235*
32	TriviaQA	Q→Q	0.838	96	CUAD	Q→Q	0.830
33	SST-2	C→C	0.998	97	GSM8K	C→C	0.977
34	SST-2	C→G	0.239*	98	GSM8K	C→G	0.831
35	SST-2	C→D	0.264*	99	GSM8K	C→D	0.728
36	SST-2	C→Q	0.501	100	GSM8K	C→Q	0.552
37	SST-2	G→C	0.861	101	GSM8K	G→C	0.844
38	SST-2	G→G	0.991	102	GSM8K	G→G	0.976
39	SST-2	G→D	0.991	103	GSM8K	G→D	0.615
40	SST-2	G→Q	0.941	104	GSM8K	G→Q	0.572
41	SST-2	D→C	0.830	105	GSM8K	D→C	0.833
42	SST-2	D→G	0.968	106	GSM8K	D→G	0.867
43	SST-2	D→D	1.000	107	GSM8K	D→D	0.793
44	SST-2	D→Q	0.929	108	GSM8K	D→Q	0.544
45	SST-2	Q→C	0.965	109	GSM8K	Q→C	0.846
46	SST-2	Q→G	0.495*	110	GSM8K	Q→G	0.814
47	SST-2	Q→D	0.638	111	GSM8K	Q→D	0.680
48	SST-2	Q→Q	0.793	112	GSM8K	Q→Q	0.755
49	MNLI	C→C	0.983	113	CNN/DM	C→C	1.000
50	MNLI	C→G	0.826	114	CNN/DM	C→G	0.571
51	MNLI	C→D	0.589	115	CNN/DM	C→D	0.595
52	MNLI	C→Q	0.654	116	CNN/DM	C→Q	0.896
53	MNLI	G→C	0.850	117	CNN/DM	G→C	0.999
54	MNLI	G→G	0.964	118	CNN/DM	G→G	0.797
55	MNLI	G→D	0.712	119	CNN/DM	G→D	0.759
56	MNLI	G→Q	0.784	120	CNN/DM	G→Q	0.927
57	MNLI	D→C	0.906	121	CNN/DM	D→C	0.992
58	MNLI	D→G	0.874	122	CNN/DM	D→G	0.651
59	MNLI	D→D	0.925	123	CNN/DM	D→D	0.783
60	MNLI	D→Q	0.831	124	CNN/DM	D→Q	0.500*
61	MNLI	Q→C	0.908	125	CNN/DM	Q→C	0.995
62	MNLI	Q→G	0.839	126	CNN/DM	Q→G	0.449*
63	MNLI	Q→D	0.692	127	CNN/DM	Q→D	0.474*
64	MNLI	Q→Q	0.836	128	CNN/DM	Q→Q	0.988

Table 9: All 128 scenarios. C=Claude, G=GPT, D=DeepSeek, Q=Qwen. *Failure (AUC $\leq$ 0.50): 11 total, in CUAD (5), SST-2 (3), CNN/DM (3).